- Increase green cover alongside roads (planting avenue trees)
- Promoting Swachh Bharat Abhiyan
- Enactment and Enforcement of stricter environmental laws
- Maintenance of air standards by proper enforcement and monitoring




Average human consumption of Oxygen per day = 550 L

Cost of 2.75 L Oxygen cylinder = ₹ 6500

Cost of 550 L of oxygen from tree = ₹ 13,00,000

---

Oxygen production by one healthy tree per year =1,00,375 L

Cost of 2.75 L oxygen cylinder = ₹ 6500

Cost of 1,00,375 L of oxygen from one tree /year = ₹ 23,72,50,000

- Reducing carbon emissions
- Encourage use of renewable energy
- Limiting the sale of firecrackers and developing eco-friendly crackers
- Make Environmental Impact Assessment mandatory

**Air Quality Index** (AQI) is a number used by government agencies to communicate to the public how polluted the air is at a given time.

| Air Quality Index | | |
|---|---|---|
| **AQI** | **Air Pollution Level** | **Colour** |
| 0-50 | Good | |
| 51-100 | Moderate | |
| 101-150 | Unhealthy for Sensitive Groups | |
| 151-200 | Unhealthy | |
| 201-300 | Very Unhealthy | |
| 301+ | Hazardous | |

## 13.3 Water Pollution

### 13.3.1 Quality of Water

Water is essential for life and for the health of the environment. As a valuable natural resource, it comprises marine, estuarine, freshwater (river and lakes) and groundwater environments that stretch across coastal and inland areas. Water has two dimensions that are closely linked: **quantity** and **quality**. Water quality is commonly defined by its physical, chemical, biological and aesthetic (appearance and smell) characteristics. A healthy environment is one in which the water quality supports a rich and varied community of organisms and protects public health.

### 13.3.2 Water Pollution

Water pollution occurs when there is a change in the chemical, physical or biological quality of water that has harmful effect(s) on living organisms that consume it or live in it.

Water pollution adversely affects water bodies due to the large amounts of natural or man-made materials let into it. When it becomes unfit for its intended use, water is considered polluted.

### 13.3.3 Sources of Water Pollution

Even though water bodies or sources can be polluted by natural causes, water pollution is usually caused by human activities. There are three main types of sources: point sources, non-point sources, leaks and spills.

**Point sources:** Discharge of pollutants at specific locations through pipelines or sewers into the water body. Factory effluents, sewage, underground mines, oil wells, oil tankers and agriculture are common point sources **(Fig. 13.2 a)**.

**Non-point sources:** Sources that cannot be traced to a single site of discharge like acid rain, dumping of the plastics in water bodies, agriculture chemical run off are common examples **(Fig. 13.2 b)**.



XII Std Zoology Chapter 13 EM.indd 223 23/12/2022 16:53:52